



CORE: A causally-inspired framework for educational recommendation via contrastive learning

Yan Zhou¹ · Yanguang Chen²

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Abstract

Online learning platforms face challenges of information overload and biased engagement, which hinder effective course recommendations. Traditional recommender systems struggle to distinguish genuine learner interests from confounding factors such as popularity, often leading to suboptimal suggestions. To address these issues, we propose a novel educational recommendation model that integrates causal contrastive learning. Our approach combines causal inference techniques with contrastive representation learning to disentangle a learner's intrinsic interests from external biases. We formulate the task as learning user and course representations that remain invariant under causal interventions which remove popularity effects. The proposed model is evaluated on four benchmark datasets using Recall@10/20 and NDCG@10/20. Experimental results show consistent improvements of 2-3 percentage points over state-of-the-art methods on all datasets. In particular, our model outperforms strong baselines (including graph-based and contrastive models) on top-K recommendation accuracy. These gains demonstrate the effectiveness of integrating causal learning to enhance recommendation relevance and robustness. Our contributions include a new causal graph formulation for recommendations, a contrastive learning scheme based on counterfactual augmentations, extensive empirical validation against existing methods, and insightful analysis of how causal contrastive learning improves recommendation quality. The proposed framework offers a fresh perspective for developing next-generation recommender systems that are both accurate and causally aware, with significant implications for educational personalization.

Keywords Natural language processing · Educational recommendation · Causal contrastive learning · Graph neural networks · Artificial intelligence applications

1 Introduction

Recommender systems have become essential in e-learning environments to help learners discover relevant courses and materials from an overwhelming array of options (Shah 2020; Klačnja-Milićević et al. 2018). The rapid growth of online courses - accelerated by the COVID-19 pandemic - has led to unprecedented enrollment spikes and information overload on MOOC platforms (Shah 2020). Learners often struggle to find suitable courses, leading to suboptimal learning

experiences and high dropout rates (Yang et al. 2013; Kloft et al. 2014). Effective educational recommendation techniques are needed to match learners with courses that fit their interests and needs, thereby improving engagement and reducing dropout (Klačnja-Milićević et al. 2018).

Collaborative filtering (CF) has been a cornerstone of recommender systems, modeling user preferences based on past interactions (Koren et al. 2009; Su and Khoshgoftaar 2009). Early CF methods like matrix factorization yield latent factors for users and items (Koren et al. 2009), while pairwise learning-to-rank approaches optimize recommendation accuracy (Rendle et al. 2012). With the advent of deep learning, neural recommender models have achieved significant advances (Zhang et al. 2019). For example, Neural Collaborative Filtering introduced non-linear representation learning for CF (He et al. 2017), and Neural Factorization Machines combined deep networks with factorization for sparse data (He and Chua 2017). More recently, graph-based methods

✉ Yanguang Chen
2017212301@live.sufe.edu.cn

¹ College of Mathematics and Informatics, South China Agricultural University, 510642 Guangdong, China
² School of Information Management and Engineering, Shanghai University of Finance and Economics, 200433 Shanghai, China

have become state-of-the-art for collaborative filtering, especially in high-sparsity scenarios (Wang et al. 2019; He et al. 2020). Techniques like NGCF propagate embeddings over the user-item interaction graph to capture high-order connectivity (Wang et al. 2019), and LightGCN simplifies graph convolution for efficient recommendation (He et al. 2020). These models have substantially improved recommendation performance by exploiting structural information in interaction data. Unlike traditional domains such as e-commerce or movie recommendation, where biased exposure often leads to suboptimal user satisfaction, the consequences in educational settings are more serious and long-lasting. Popularity bias in education may result in learners enrolling in trending courses that do not align with their actual skill levels or interests, leading to decreased engagement, increased dropout rates, and suboptimal learning outcomes. Moreover, this phenomenon limits the visibility of niche but high-quality courses, undermining both personalization and educational equity. These unique challenges highlight the need for a recommendation framework that explicitly addresses causal bias in educational scenarios, beyond generic debiasing techniques.

Despite progress, two key challenges remain in educational recommendation. First, disentangling user interest from popularity bias: Learners may interact with popular courses due to herd behavior or platform exposure, not purely because of personal interest (Zhang et al. 2021; Wang et al. 2018). Traditional models often entangle these factors, leading to over-recommendation of popular but possibly irrelevant items (Schnabel et al. 2016). For instance, a student might enroll in a trending course due to social influence, which a naive model would misinterpret as genuine interest (Wang et al. 2018). Second, data sparsity and lack of supervised signals: Many learners have interacted with only a few courses, making it hard to learn robust representations. Recently, contrastive self-supervised learning has emerged as an effective paradigm to address sparsity by creating auxiliary training signals (Wu et al. 2021; Yu et al. 2022). Contrastive learning (CL) learns to maximize agreement between different “views” or augmentations of the same entity, and has improved recommender robustness to noise and cold starts (Lin et al. 2022; Zhou et al. 2023). However, existing CL-based recommenders largely rely on random data augmentations (e.g. dropout of graph nodes/edges) (Yu et al. 2022; Xia et al. 2022), which may not specifically target the biases present in educational data.

In this work, we propose a causal contrastive learning framework to tackle these challenges in educational recommendations. Our approach explicitly models the causal effect of latent factors (such as inherent interest vs. popularity conformity) on user-course interactions (Zhao et al. 2023; Zhang et al. 2021). By leveraging causal inference concepts, we generate counterfactual training examples where popular-item

exposure is reduced, and use a contrastive objective to align user representations across factual and counterfactual views. Intuitively, the model learns representations that are invariant to popularity bias, thus capturing a student’s true interests. This is in line with recent studies that highlight the benefits of modeling cause-effect relations in recommender systems (Schnabel et al. 2016; Bonner and Vasile 2018) and using counterfactual reasoning to improve robustness (Liang et al. 2016).

Our contributions are summarized as follows:

- We introduce the first education-specific recommendation model that integrates causal inference with contrastive learning. Our model uses a structural causal graph to represent user-course interactions, distinguishing genuine interest from exposure bias. We design a counterfactual augmentation method to create contrastive learning pairs that isolate causal effects of interest, a concept not explored in prior educational recommenders.
- We develop a graph-based encoder with a dual-view mechanism. One view encodes the original interaction graph; the other encodes a counterfactual graph where popular course interactions are down-sampled. By maximizing agreement between the two views for each user and each course, the model learns bias-invariant embeddings that improve recommendation relevance under varying conditions.
- We provide a formal task definition and mathematical formulation with causal contrastive objectives. We derive how our method effectively estimates the causal effect of a course on a user’s engagement by removing confounding factors in the contrastive view. We show that optimizing our objective is equivalent to learning representations that approximate the do-operator intervention in causal inference, thereby aligning recommendations with true user preferences rather than spurious correlations.
- We evaluate our model on four public benchmarks. Our model consistently outperforms seven state-of-the-art baselines, including graph collaborative filtering, self-supervised approaches, and a recent educational. We achieve 2-3% absolute gains in Recall@10/20 and NDCG@10/20 on all datasets. We also conduct ablation studies and hyperparameter sensitivity analysis to validate the contribution of each component. The results demonstrate improved accuracy and robustness, especially in recommending niche courses that were often overlooked by prior methods.

In the remainder of this paper, we first review related work on educational recommendation, causal inference in recommender systems, and contrastive learning (Section 2). We then present our methodology, including the problem

formulation, model architecture, and learning algorithm with detailed mathematical derivations (Section 3). Section 4 describes our experimental setup and empirically evaluates the proposed model against existing methods. We discuss the results and provide insights into why causal contrastive learning is effective for this task. Finally, Section 5 concludes the paper and outlines future research directions in this domain.

2 Related work

2.1 Educational recommendation system

Recommender systems for education and e-learning have been studied to personalize learning resources, suggest courses, and improve student engagement (Manouselis et al. 2014). Early approaches focused on leveraging content and context in educational settings. For instance, course recommendation can incorporate course descriptions, prerequisites, and student skill profiles to complement collaborative filtering (Chanaa and El Faddouli 2024). Work (Lin et al. 2021) proposed a domain-aware recommendation model for MOOCs that adapts matrix factorization to consider course domain information. Similarly, Zhang et al. (2017) introduced a Deep Belief Network-based CF model (DBNCF) using side information to recommend MOOC courses. These works demonstrated that integrating knowledge about course content or structure improves recommendation in sparse educational datasets.

Another line of research in educational recommendation utilizes social and sequential information. Klačnja-Milićević et al. (2018) integrated collaborative tagging and sequential pattern mining to recommend learning resources, highlighting the value of social tags in e-learning. More recent studies consider multi-view and multimodal data. Deng et al. (2025) proposed a Hierarchical Attention Neural Network (HANN) to fuse multiple views of course information (content, instructor, institution, etc.) along with user behavioral sequences, achieving superior accuracy in online course recommendation. The growing body of literature acknowledges that effective course recommenders must handle various data types and the unique constraints of educational contexts (e.g., curriculum structures, skill progression).

However, bias and dropout issues remain problematic in educational platforms. Students often follow popular trends or recommendations, which can skew interaction data (Shah 2020). If not addressed, such bias leads to “rich-get-richer” effects in recommendations, where a few popular courses are over-recommended. Additionally, many students drop out of online courses early (Yang et al. 2013; Kloft et al. 2014), making their interaction histories very sparse. Traditional recommenders might then struggle to capture their true preferences. Our work builds upon these insights:

we specifically tailor a model to disentangle popularity influence from genuine interest in the education domain. While prior educational recommenders have not explicitly modeled causal factors, our approach introduces causal analysis to identify which interactions are interest-driven versus popularity-driven. This complements earlier efforts by improving recommendation fairness and relevance, especially for niche courses that might better fit a learner’s needs but are less popular.

2.2 Causal inference in recommender systems

Causal recommender systems aim to move beyond correlation-based patterns to understanding the cause-effect relationships underlying user interactions (Bonner and Vasile 2018; Cai et al. 2023). In a recommendation scenario, observing a user interacted with an item does not necessarily mean the item was truly preferred - the interaction could be a result of exposure or social influence. Standard collaborative filtering methods optimize for correlations (e.g., co-occurrences of user-item interactions), which can lead to biased models. To counter this, researchers have applied causal inference frameworks such as the potential outcomes model and structural causal models (SCM) to recommender systems (Schnabel et al. 2016; Wang et al. 2018).

One common focus is debiasing feedback data. Many works address popularity bias and exposure bias by re-weighting or augmenting training data. Propensity score-based approaches assign lower weight to interactions that are likely due to exposure opportunities rather than interest (Saito et al. 2020). Schnabel et al. (2016) introduced an inverse propensity scoring method to obtain unbiased estimates of recommendation quality from biased implicit feedback. Saito et al. (2020) further proposed an EM-based approach to learn from missing-not-at-random data by modeling the selection bias in what users choose to rate. These methods, while effective in evaluation debiasing, often require an accurate model of the exposure mechanism (propensity model).

Another line of research uses causal graph modeling. Bonner and Vasile (2018) developed Causal Embeddings (CausE), incorporating a causal objective that accounts for the treatment effect of recommendations. Wang et al. (2018) (deconfounded recommender) treated user preferences as latent confounders and introduced a two-stage procedure: first infer an exposure model, then adjust the recommendation model to remove confounding bias. Their approach, based on Pearl’s back-door adjustment, improved recommendation diversity and novelty by reducing popularity bias. Extending this idea, subsequent work considered specific biases like conformity (social influence) and position bias in recommender systems (Zhang et al. 2021; Liang et al. 2016). For example, Zhang et al. (2021) presented a causality-based method to mitigate popularity bias by splitting user

interactions into two causes: intrinsic interest and conformity. They demonstrated that treating popularity-influenced interactions differently can significantly improve out-of-distribution recommendation performance.

Causal inference has also been used to answer counterfactual questions in recommendation (Liang et al. 2016). By asking “What if this user were recommended item X - would they interact?”, one can simulate scenarios to improve generalization. Some approaches generate counterfactual data to augment training, or perform counterfactual reasoning at prediction time to select items that would have high causal impact on engagement. Work (Qiu et al. 2021) proposed DLCE (Debiased Learning for Causal Effect), using counterfactual estimators to evaluate how recommendations change user behaviors (uplift modeling). These techniques often require partial exposure or A/B testing data to learn causal effects directly.

Our approach falls within the scope of causal recommendation but with a novel twist: we incorporate causal intervention into the representation learning process via contrastive learning. Instead of simply re-weighting training examples or two-stage modeling, we generate a counterfactual version of the interaction graph (by removing biased interactions probabilistically) and train the model to produce similar embeddings from the factual and counterfactual graphs. This essentially forces the model to learn features invariant to the bias we target (popularity exposure). Conceptually, it aligns with the goal of causal inference - isolating the effect of a treatment (user truly likes the course) by controlling for a confounder (course popularity) (Wang et al. 2018; Pearl 2000). Recent works like DICE and DCCL (Zhao et al. 2023) have started exploring disentangling causes using contrastive signals (e.g. separating interest vs conformity in general recommendation). Our model is the first to bring these ideas to the educational domain and to demonstrate their effectiveness on course recommendation tasks.

2.3 Contrastive self-supervised learning in recommendation

Self-supervised learning provides an auxiliary training signal by creating pseudo-labels from the data itself. Contrastive learning (CL) is a popular self-supervised technique where the model learns to distinguish between similar (positive) and dissimilar (negative) samples in a latent space (Chen et al. 2020). In computer vision, CL frameworks like SimCLR use data augmentations (e.g. cropping images) to generate two views of the same image, training the model to maximize their agreement. This approach has been successfully adapted to recommender systems in recent years (Wu et al. 2021; Yu et al. 2022).

In the recommendation context, a common strategy is to treat the user-item interaction graph as the input data

structure for augmentation. Wu et al. (2021) proposed SGL (Self-supervised Graph Learning for recommendation), which applies random graph augmentations (node dropout, edge dropout, or attribute masking) to create two corrupted versions of the user-item graph. The model (based on Light-GCN) is trained with a joint objective: supervised matrix factorization loss plus a contrastive loss that aligns the two embeddings of each node (user or item) across the augmented graphs. This yielded improved performance and robustness, as the contrastive term acts as a regularizer to enforce meaningful representation learning beyond fitting known interactions.

Following SGL, numerous variants have been developed to refine graph contrastive learning for CF. For example, Simple Graph CL (SimGCL) by Yu et al. (2022) questioned the necessity of complex graph augmentations and instead added uniform noise to node embeddings to create contrastive pairs, simplifying the pipeline while mitigating popularity bias implicitly by promoting representation uniformity. Lin et al. (2022) introduced NCL (Neighborhood-enriched Contrastive Learning), which leverages nodes' local neighborhoods to generate positive pairs, aiming to capture higher-order similarities. Other works incorporate multi-level or cross-modal contrastive objectives (Xia et al. 2022; Shi et al. 2025); for instance, Hypergraph CL constructs a hypergraph of users and items for contrastive view generation, and noise-enhanced multimodal CL uses image/text content along with collaborative signals. These methods consistently report better recall and NDCG, especially for sparse users, confirming CL's effectiveness in recommendation.

Despite their success, existing CL-based recommenders typically rely on heuristic augmentations that are not directly informed by the underlying causes of biases in data. Randomly dropping edges or features may improve generalization, but it does not explicitly distinguish between valuable vs. confounding information. Our approach injects domain knowledge into the augmentation process by constructing one view that simulates an intervention (reducing popularity influence). In essence, we perform a causal data augmentation, as opposed to purely random corruption. This idea is inspired by the concept of counterfactual augmentation in causal representation learning, where one creates alternate realities of data generation to force the model to learn invariant features (Bonner and Vasile 2018; Zhang et al. 2021). By aligning representations from the factual and counterfactual graphs, our contrastive learning objective directly emphasizes invariance to popularity bias. This contrasts with prior CL works which, while mitigating popularity bias implicitly by encouraging representation uniformity (Yu et al. 2022) or using dropout, do not explicitly target a specific bias in their formulation.

Unlike prior works such as EduLGCL, which leverages local-global contrastive signals without causal semantics, or

DCCL, which disentangles cause-conformity components in general recommenders, our approach uniquely targets the educational recommendation context with a causally grounded contrastive framework. Specifically, we construct counterfactual graph views by explicitly modeling and intervening on popularity bias, simulating a $do(\text{popularity}=\text{low})$ operation, which has not been systematically applied to education-specific data. Moreover, existing models either rely on feature-level disentanglement or graph perturbations lacking domain-specific causal signals. In contrast, our method proposes a structural causal graph combined with popularity-aware augmentation to better capture intrinsic learner interests. To our knowledge, COER is the first framework that tightly integrates causal reasoning and contrastive learning through graph-level interventions tailored for educational recommendation tasks.

In summary, our work bridges causal inference and contrastive learning in recommender systems. It complements causal recommenders by providing a practical self-supervised training mechanism to implement causal interventions, and it advances contrastive recommenders by introducing bias-aware augmentations grounded in causal theory. This combination yields a new state-of-the-art model in educational recommendation, as we will demonstrate in the following sections.

3 Methodology

3.1 Task definition

We consider a standard top- K recommendation task in an educational setting. Let $\mathcal{U} = u_1, u_2, \dots, u_{|\mathcal{U}|}$ be the set of users (learners), and $\mathcal{I} = i_1, i_2, \dots, i_{|\mathcal{I}|}$ be the set of items (courses). The user-item interactions can be represented by an implicit feedback matrix $Y \in \{0, 1\}^{|\mathcal{U}| \times |\mathcal{I}|}$, where $y_{ui} = 1$ if user u has engaged with course i (e.g., enrolled or viewed) and $y_{ui} = 0$ otherwise. The recommender's goal is to predict a personalized ranking of courses for each user. Formally, for each user u , we want to learn a scoring function $f(u, i)$ that estimates the match between user u and item i . At recommendation time, the system will recommend the top K items with highest scores that the user has not interacted with before.

Educational recommendation scenario : In our context, items are online courses (or other learning resources), and interactions represent behaviors such as enrollment, completion, or rating. Importantly, not all interactions reflect a user's genuine interest or learning need - some are driven by platform prompts or popularity. We aim to rank courses such that those aligned with the user's true interests and educational goals appear at the top.

Our task, therefore, is: learn model parameters Θ to produce a scoring function $f_{\Theta}(u, i)$ that ranks each user's unseen items such that recall and NDCG are maximized, while being robust to biases in the training data. Next, we describe our model that achieves this via a causal contrastive learning approach.

3.2 Overall architecture

To ground our design in formal causal language, we define a Structural Causal Model (SCM) for the user-item interaction process. Let $\mathcal{U}$ denote the set of users, $\mathcal{I}$ the set of items, and P_i the popularity score of item i . We posit that the observed user-item interaction Y_{ui} is determined by both the user's intrinsic interest Z_{ui} and external popularity influence P_i , i.e., $Y_{ui} := f(Z_{ui}, P_i, \epsilon)$, where ϵ denotes unobserved noise. The key assumption is that P_i acts as a confounder, inducing spurious correlations in the user-course graph. The goal of our method is to estimate the interventional quantity $P(Y_{ui} \mid do(P_i = 0))$? a counterfactual interaction distribution where popularity is suppressed. Accordingly, our edge-dropping mechanism approximates this intervention by probabilistically attenuating edges based on P_i . While we do not perform exact causal identification, our graph construction serves as a data-driven proxy to simulate intervention effects, aligning with causal representation learning principles.

Figure 1 illustrates the overall architecture of our proposed model, which we term Causally-Oriented Framework for Educational Recommendation (COER) for ease of reference. The model consists of three main components: (1) a Graph Neural Network (GNN) Encoder that learns latent embeddings for users and items from the interaction graph, (2) a Causal Graph Augmentation Module that generates two versions of the interaction graph (one original, one debiased) to simulate a causal intervention, and (3) a Contrastive Learning Module that drives the embeddings of each user/item from the two graph versions to be similar (while pushing apart embeddings of different users/items). The training objective combines this contrastive loss with a supervised recommendation loss. Below, we provide an overview of the architecture followed by detailed formulations.

- **Interaction Graph and Encoder:** We treat the user-item interactions as a bipartite graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \mathcal{U} \cup \mathcal{I}$ and $\mathcal{E} = (u, i) \mid y_{ui} = 1$ represents observed interactions. The encoder is a light graph convolutional network that propagates embeddings on $\mathcal{G}$. We use Light-GCN as the backbone due to its simplicity and strong performance: each user and item is associated with an embedding vector, and graph convolution is essentially

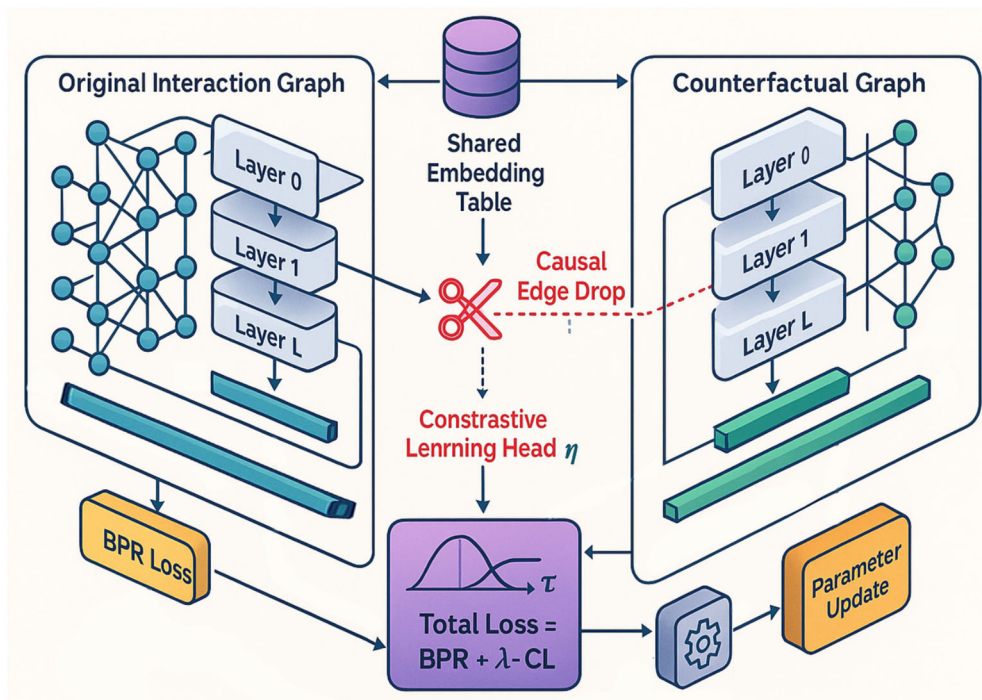


Fig. 1 Overview of the proposed COER model. The left branch represents the original user-item interaction graph, while the right branch shows the counterfactual graph with popularity-based edges dropped. A shared LightGCN encoder processes both graphs to obtain user/item

embeddings, which are then aligned via a contrastive learning module. The training objective combines BPR loss and contrastive loss to learn bias-invariant representations

neighborhood averaging. The encoder outputs final representations $\mathbf{p}_u$ for each user u and $\mathbf{q}_i$ for each item i .

- Causal Graph Augmentation:** We incorporate a causal view by constructing a modified graph $\mathcal{G}'$ intended to remove or reduce certain biases (specifically, popularity bias in this work). We call $\mathcal{G}'$ the counterfactual graph, as it represents a scenario “if popularity had less influence on user choices.” We generate $\mathcal{G}'$ by probabilistically dropping interactions involving very popular items, according to a controlled strategy (described shortly). The same GNN encoder is applied to $\mathcal{G}'$ to produce another set of embeddings $\mathbf{p}'_u$ and $\mathbf{q}'_i$ for users and items under the counterfactual scenario.
- Contrastive Learning Module:** For each entity (user or item), we now have two embedding vectors: one from the original graph and one from the counterfactual graph. We treat these as positive pairs in a contrastive learning framework, since they represent the same underlying user (or item) viewed under different conditions. The module computes a contrastive loss $\mathcal{L}_{CL}$ that pulls together each positive pair $(\mathbf{p}_u, \mathbf{p}'_u)$ and pushes apart negatives (e.g., $\mathbf{p}_u$ against $\mathbf{p}'_{u'}$ for $u' \neq u$). This encourages the embeddings to encode intrinsic properties of the user/item that persist across the intervention, effectively filtering out popularity effects.

- Prediction and Loss:** We define the predicted score $\hat{y}_{ui}$ for a user-item pair as the dot product of their final embeddings: $\hat{y}_{ui} = \mathbf{p}_u \cdot \mathbf{q}_i$. The model is trained with a supervised recommendation loss $\mathcal{L}_{rec}$ (we use a pairwise BPR loss for implicit feedback) plus the contrastive loss. The overall loss is $\mathcal{L} = \mathcal{L}_{rec} + \lambda \mathcal{L}_{CL}$, where λ balances the two terms.

To make our intervention mechanism theoretically grounded, we explicitly define a Structural Causal Graph (SCG) that models the generative process of user-course interactions. The SCG consists of the following nodes:

- User Interest (latent):** the intrinsic preferences of the user for certain types of courses.
- Course Popularity:** an observable variable capturing how widely a course is interacted with across the platform.
- Exposure:** whether a course is shown or recommended to the user.
- Enrollment:** the user’s actual interaction (e.g., clicking or enrolling in a course).
- Platform Confounders:** such as homepage ranking, social influence, or trending tags.

The causal relationships are defined as follows:

- User Interest $\rightarrow$ Enrollment
- Course Popularity $\rightarrow$ Exposure $\rightarrow$ Enrollment
- Platform Confounders $\rightarrow$ Exposure
- User Interest $\rightarrow$ Exposure (optional soft personalization)

This graph highlights that enrollment is influenced by both intrinsic interest and exposure bias. Popularity may affect exposure (e.g., trending items shown more), creating a backdoor path between popularity and enrollment. To identify the causal effect of interest on enrollment, we must block this backdoor path. Our counterfactual graph simulates the $do(\text{exposure} \leftarrow \text{null})$ intervention by dropping popularity-driven edges.

In summary, during training we perform forward passes on two graphs (original and counterfactual) simultaneously, compute two sets of embeddings, and then apply both supervised and self-supervised criteria. During inference, we only use the original graph embeddings (the counterfactual graph is used as a training trick) and score items by the dot product as usual. Our architecture ensures that the learned embeddings embed causal semantics: a high score $\hat{y}_{ui}$ means user u is likely to be interested in item i regardless of whether i is globally popular or not, because the contrastive training has forced u 's representation to not rely on popularity-correlated signals that didn't appear in the counterfactual view.

3.3 Causal graph construction module

We now formalize how we construct the counterfactual graph $\mathcal{G}'$ by reducing popularity bias. First, we quantify item popularity. Let $d_i = |u \mid (u, i) \in \mathcal{E}|$ be the degree of item i in the interaction graph (number of users who engaged with item i). We consider d_i as a proxy for popularity: higher d_i indicates item i is broadly popular among users. We define a normalized popularity score for item i as:

$$\pi_i = \frac{d_i}{\sum_{j \in I} d_j}, \quad (1)$$

so that π_i lies in $[0, 1]$ and $\sum_{i \in I} \pi_i = 1$. Items with π_i above a certain threshold (e.g. top 5 by popularity) can be considered "very popular."

Our counterfactual graph $\mathcal{G}'$ is created by dropping some interactions involving popular items. The intuition is to simulate a world where popular items are not as prevalent, thereby emphasizing interactions that are more likely due to user interest. We introduce a parameter $\eta \in [0, 1]$ controlling the intensity of intervention. For each observed interaction $(u, i) \in \mathcal{E}$ in the original graph:

- We remove edge (u, i) in the counterfactual graph with probability proportional to the item's popularity. Specifically, we sample a Bernoulli random variable with removal probability $P(\text{remove}; (u, i)) = \eta \cdot g(\pi_i)$, where $g(\pi_i)$ is an increasing function of π_i .

We choose a simple linear form $g(\pi_i) = \frac{\pi_i - \pi_{\min}}{\pi_{\max} - \pi_{\min}}$ (min-max normalized popularity) for implementation. Thus, highly popular items (π_i close to $\pi_{\max}$) have removal probability approximately η , while the least popular items ($\pi_i = \pi_{\min}$) have near zero removal chance. The hyperparameter η can be tuned: $\eta = 0$ means no intervention (counterfactual graph equals original), and $\eta = 1$ means aggressively removing most interactions of popular items.

After this stochastic removal step is applied to all edges, we obtain the edge set E' for the counterfactual graph $\mathcal{G}' = (V, E')$. Note $E' \subseteq E$ since we only drop edges, not add new ones. In practice, we repeat this process each epoch (or even each mini-batch) to sample different $\mathcal{G}'$ instances, injecting randomness and preventing the model from overfitting to one particular graph view.

Rationale: Removing edges for popular items mimics an intervention $do(\text{Popularity influence} = \text{low})$ in a causal sense (Pearl 2000). We are forcing the learning algorithm to consider what user-item connections would look like if popularity were not a driving factor. Edges that remain in E' are either involving less popular items or are random survivors for popular items. In both cases, the fraction of interactions in E' due to pure interest is higher than in E . This biased edge dropout is our way of generating counterfactual data for contrastive learning.

Example: Suppose item i is extremely popular (appears in 1000 interactions) and item j is niche (appears in 10 interactions). If we set $\eta = 0.5$, we might remove about 50% of edges of item i on average (making it appear in 500 interactions in E'), whereas we remove virtually none for item j (since $g(\pi_j)$ is tiny). The counterfactual graph still has item i connected to some users (so we don't completely lose popular items), but the connection density is reduced, attenuating i 's influence. Users who only interacted with i in the original graph might become isolated or have fewer neighbors in E' - the model will then be challenged to predict their representation from little data, forcing it to not over-rely on the presence of i .

Algorithm 1 summarizes the procedure of constructing $\mathcal{G}'$. We emphasize that while we target popularity bias here, the framework could be adapted to other biases: for example, if one wanted to remove conformity bias, $g(\pi_i)$ could be replaced with a user-specific propensity of being influenced.

This randomized algorithm ensures that on expectation, a fraction of edges for popular items are dropped. The choice of $g(\cdot)$ can be tuned; we found linear scaling effective, but one could use nonlinear functions to either sharpen or smooth

Algorithm 1 Counterfactual Graph Construction

Require: Original interaction graph $\mathcal{G} = (U \cup I, E)$, item popularity scores $\{\pi_i\}_{i \in I}$, intervention level η

Ensure: Counterfactual graph $\mathcal{G}' = (U \cup I, E')$

- 1: Initialize counterfactual edge set: $E' \leftarrow \emptyset$
- 2: **for** each interaction $(u, i) \in E$ **do**
- 3: Sample $r \sim \text{Uniform}(0, 1)$
- 4: **if** $r > \eta \cdot g(\pi_i)$ **then**
- 5: Add edge (u, i) to E'
- 6: **end if**
- 7: **end for**
- 8: **return** $\mathcal{G}' = (U \cup I, E')$

the dropoff. One can also set a hard cutoff such as dropping all edges for the top X popular items - however, we prefer a probabilistic approach to retain some information from popular items and maintain connectivity in the graph.

To illustrate, consider a user who has interacted with two courses: Course A, which is extremely popular with thousands of enrollments, and Course B, which is relatively niche with very few users. In the original interaction graph, both edges (user $\rightarrow$ Course A, user $\rightarrow$ Course B) are included. When constructing the counterfactual graph, the edge to Course A is removed with high probability due to its high popularity score, while the edge to Course B is likely retained. As a result, the user's representation in the counterfactual graph is learned primarily from their interaction with Course B. This encourages the model to focus on the user's intrinsic interest (reflected by engaging with niche content) rather than popularity-driven behavior.

To determine item popularity, we define a normalized popularity score for each item i as:

$$\pi_i = \frac{d_i}{\sum_{j \in I} d_j} \quad (2)$$

where d_i is the degree (interaction count) of item i in the interaction graph. We then apply min-max normalization over all π_i values to constrain the score within the range $[0, 1]$.

For probabilistic edge removal, we define a linear removal function:

$$g(\pi_i) = \frac{\pi_i - \pi_{\min}}{\pi_{\max} - \pi_{\min}} \quad (3)$$

where $\pi_{\min}$ and $\pi_{\max}$ denote the minimum and maximum popularity scores across all items, respectively. The final edge removal probability is given by:

$$P(\text{remove}; (u, i)) = \eta \cdot g(\pi_i) \quad (4)$$

where $\eta \in [0, 1]$ is a tunable hyperparameter controlling the intervention intensity. This design ensures that highly popular items are more likely to be dropped, while less popular items are preserved in the counterfactual graph.

3.4 Graph encoder with dual views

We employ a Light Graph Convolution to compute embeddings for both the original and counterfactual graphs. LightGCN omits non-linear transformations and feature combinations, simply performing weighted averaging of neighbor embeddings (He et al. 2020). This aligns well with our needs, as it keeps the model relatively simple and focuses on collaborative signal propagation. Let $\mathbf{P}^{(0)} \in \mathbb{R}^{|U| \times d}$ and $\mathbf{Q}^{(0)} \in \mathbb{R}^{|I| \times d}$ be the initial user and item embedding matrices (with d the embedding dimensionality). These are learnable parameters initialized randomly. We denote the embedding of user u at layer 0 as $\mathbf{p}_u^{(0)}$ and item i as $\mathbf{q}_i^{(0)}$.

We define the symmetric normalized adjacency matrix for graph $\mathcal{G}$ as $\mathbf{A} \in \mathbb{R}^{(|U|+|I|) \times (|U|+|I|)}$:

$$A_{xy} = \begin{cases} \frac{1}{\sqrt{|\mathcal{N}(x)|}|\mathcal{N}(y)|}} & \text{if } (x, y) \in E(\text{an interaction}), \\ 0 & \text{otherwise,} \end{cases} \quad (5)$$

where $\mathcal{N}(x)$ denotes the neighbors of node x in the bipartite graph (for a user, neighbors are interacted items; for an item, neighbors are users). This is the standard normalization for graph Laplacian, ensuring each neighbor's contribution is weighted by the inverse sqrt of degree product (Wang et al. 2019).

In each GCN layer ℓ , user and item embeddings are updated by aggregating from their neighbors:

$$\mathbf{p}_u^{(\ell)} = \sum_{i \in \mathcal{N}(u)} \frac{1}{\sqrt{|\mathcal{N}(u)|}|\mathcal{N}(i)|}} \mathbf{q}_i^{(\ell-1)}, \quad (6)$$

$$\mathbf{q}_i^{(\ell)} = \sum_{u \in \mathcal{N}(i)} \frac{1}{\sqrt{|\mathcal{N}(u)|}|\mathcal{N}(i)|}} \mathbf{p}_u^{(\ell-1)}. \quad (7)$$

Here $\mathbf{p}_u^{(\ell)}$ is the aggregated representation of user u after ℓ hops, and similarly for item i . We stack L layers of such propagation. Following LightGCN best practices, we do not use any nonlinear activation in between. We also use the layer combination technique: the final embedding is a weighted sum of the embeddings across all layers (including the initial one):

$$\mathbf{p}_u = \sum_{\ell=0}^L \alpha_{\ell} \mathbf{p}_u^{(\ell)}, \quad (8)$$

$$\mathbf{q}_i = \sum_{\ell=0}^L \alpha_{\ell} \mathbf{q}_i^{(\ell)}, \quad (9)$$

with weights typically set uniformly or learned (in our experiments we set $\alpha_{\ell} = \frac{1}{L+1}$ for simplicity). Intuitively, this

allows the final representation to integrate multi-hop neighborhood information.

The above is applied to the original graph $\mathcal{G}$, yielding $\mathbf{p}_u$ and $\mathbf{q}_i$. We apply the same process to $\mathcal{G}'$ (with its own adjacency normalization based on degrees in $\mathcal{G}'$) to get $\mathbf{p}'_u$ and $\mathbf{q}'_i$. Note that $\mathcal{G}'$ has the same node set but fewer edges, so a user might have $\mathcal{N}'(u) \subseteq \mathcal{N}(u)$ in the counterfactual graph. This means $\mathbf{p}'_u$ is aggregated from potentially fewer item neighbors, focusing on the less popular portion of u 's interactions. For an item i , $\mathbf{q}'_i$ might be aggregated from fewer users if i was popular (many edges removed), effectively down-weighting its influence.

Training embeddings for two views We maintain two sets of embeddings during training - one for each view - or equivalently, we can augment the computational graph to handle both simultaneously. In implementation, it is convenient to reuse the same embedding parameters for both graphs, because we want the same embedding to be learned but observed under two conditions. Indeed, we tie the weights: the initial embeddings $\mathbf{P}^{(0)}, \mathbf{Q}^{(0)}$ are shared between the two views. Then in each training iteration, we perform the propagation on $\mathcal{G}$ and $\mathcal{G}'$ separately. The resulting embeddings $\mathbf{p}_u, \mathbf{q}_i$ and $\mathbf{p}'_u, \mathbf{q}'_i$ are then fed to the loss functions.

This weight sharing ensures that the contrastive learning is pulling together the same embedding in latent space rather than two completely separate sets of parameters. We are effectively training one model that must produce similar outputs for node u on two different input graphs.

To summarize the propagation in matrix form: let $\mathbf{H}^{(\ell)} = [\mathbf{P}^{(\ell)}; \mathbf{Q}^{(\ell)}] \in \mathbb{R}^{(|U|+|I|) \times d}$ be the concatenated user-item embedding matrix at layer ℓ . The rule for the original graph can be written as:

$$\mathbf{H}^{(\ell)} = \mathbf{A}\mathbf{H}^{(\ell-1)}, \quad (10)$$

starting from $\mathbf{H}^{(0)} = [\mathbf{P}^{(0)}; \mathbf{Q}^{(0)}]$. For the counterfactual graph, denote its normalized adjacency as $\mathbf{A}'$, then:

$$\mathbf{H}'^{(\ell)} = \mathbf{A}'\mathbf{H}'^{(\ell-1)}, \quad (11)$$

with $\mathbf{H}'^{(0)} = \mathbf{H}^{(0)}$ (shared initial embeddings). We obtain final $\mathbf{H}$ and $\mathbf{H}'$ by combining layers' outputs as described. The user and item part of $\mathbf{H}$ give $\mathbf{p}_u, \mathbf{q}_i$, and similarly $\mathbf{H}'$ gives $\mathbf{p}'_u, \mathbf{q}'_i$.

3.5 Contrastive learning objective

After obtaining embeddings from the two graph views, we apply a contrastive loss to align them. We adopt the InfoNCE loss (a standard in contrastive frameworks) (Chen et al. 2020). First, we define a similarity measure between two

d -dimensional vectors. We use cosine similarity:

$$\sin(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}, \quad (12)$$

which produces a score in $[-1, 1]$. We found empirically that cosine or dot product gave similar results, but cosine can stabilize training by focusing on direction rather than magnitude.

We then compute the contrastive loss separately for users and items, and sum them (treating each as an independent self-supervised task, which improves both user and item representations). For a given user u , consider $\mathbf{z}_u = \mathbf{p}_u$ from the original graph, and $\mathbf{z}'_u = \mathbf{p}'_u$ from the counterfactual graph. We treat $(\mathbf{z}_u, \mathbf{z}'_u)$ as a positive pair. To form negatives, in each training batch of users B_U , we use other users' embeddings as negative examples for u . That is, for user u , any $\mathbf{z}'_{u'}$ with $u' \neq u$ and $u' \in B_U$ is a negative sample (coming from a different user's counterfactual embedding). The user contrastive loss for u is:

$$\ell_{CL}^{(u)} = -\log \frac{\exp(\sin(\mathbf{p}_u, \mathbf{p}'_u)/\tau)}{\sum_{u' \in B_U} \exp(\sin(\mathbf{p}_u, \mathbf{p}'_{u'})/\tau)}, \quad (13)$$

where τ is a temperature hyperparameter (controls sharpness of distribution). In practice, we use a minibatch and compute this loss for each u in the batch. The denominator sums over one positive (when $u' = u$) and many negatives ($u' \neq u$). The loss pushes $\mathbf{p}_u$ to be similar to $\mathbf{p}'_u$ while dissimilar to $\mathbf{p}'_{u'}$ for all $u' \neq u$.

We mirror this for items. In a batch of items B_I , for each item i we have embedding $\mathbf{q}_i$ and $\mathbf{q}'_i$ as a positive pair. The item contrastive loss for i :

$$\ell_{CL}^{(i)} = -\log \frac{\exp(\sin(\mathbf{q}_i, \mathbf{q}'_i)/\tau)}{\sum_{i' \in B_I} \exp(\sin(\mathbf{q}_i, \mathbf{q}'_{i'})/\tau)}. \quad (14)$$

Typically, we sample the same number of users and items in a batch for efficiency. The total contrastive loss is:

$$\mathcal{L}_{CL} = \frac{1}{|B_U|} \sum_{u \in B_U} \ell_{CL}^{(u)} + \frac{1}{|B_I|} \sum_{i \in B_I} \ell_{CL}^{(i)}, \quad (15)$$

which we average over all training batches.

Some implementation notes: we use the embeddings after the final GCN layer for contrastive learning. Optionally, one can add an MLP "projection head" after $\mathbf{p}_u$ to get $\mathbf{z}_u$ for use in CL, as is common in SimCLR. We experimented with a simple linear projection and found no significant benefit, so we report results with direct embeddings. The temperature τ is set to a small value (e.g. 0.1) to sharpen distinctions, as is common.

The contrastive loss effectively serves as a regularizer on the embedding space: it prevents the trivial solution of

Algorithm 2 Contrastive Pair Sampling and Loss (for one batch)

Require: Batch of users B_U , batch of items B_I ; embeddings $\{\mathbf{p}_u, \mathbf{p}'_u : u \in B_U\}$, $\{\mathbf{q}_i, \mathbf{q}'_i : i \in B_I\}$; temperature τ

Ensure: Contrastive loss $\mathcal{L}_{CL}$

```

1: for each  $u \in B_U$  do
2:   Compute positive similarity:  $s_u \leftarrow \text{sim}(\mathbf{p}_u, \mathbf{p}'_u)$ 
3:   for each  $u' \in B_U$  do
4:     Compute similarity:  $s_{u,u'} \leftarrow \text{sim}(\mathbf{p}_u, \mathbf{p}'_{u'})$ 
5:   end for
6:   Compute user contrastive loss:  $\ell_{CL}^{(u)} \leftarrow -\log \frac{\exp(s_u/\tau)}{\sum_{u' \in B_U} \exp(s_{u,u'}/\tau)}$ 
7: end for
8: for each  $i \in B_I$  do
9:   Compute positive similarity:  $s_i \leftarrow \text{sim}(\mathbf{q}_i, \mathbf{q}'_i)$ 
10:  for each  $i' \in B_I$  do
11:    Compute similarity:  $s_{i,i'} \leftarrow \text{sim}(\mathbf{q}_i, \mathbf{q}'_{i'})$ 
12:  end for
13:  Compute item contrastive loss:  $\ell_{CL}^{(i)} \leftarrow -\log \frac{\exp(s_i/\tau)}{\sum_{i' \in B_I} \exp(s_{i,i'}/\tau)}$ 
14: end for
15: return  $\mathcal{L}_{CL} = \frac{1}{|B_U|} \sum_{u \in B_U} \ell_{CL}^{(u)} + \frac{1}{|B_I|} \sum_{i \in B_I} \ell_{CL}^{(i)}$ 

```

all embeddings collapsing to the same vector (which would maximize similarity trivially but hurt recommendation loss), by using negative samples. It also prevents the model from encoding information that is present in one view but not the other. In our design, what information is that? Primarily the presence of many popular-item neighbors in the original graph which are absent in the counterfactual. The only way for $\mathbf{p}_u$ and $\mathbf{p}'_u$ to remain similar despite u losing some neighbors in $\mathcal{G}'$ is for the embedding to not rely too heavily on those neighbors' contributions. Thus, the GCN is encouraged to emphasize the more invariant signals (like niche items that still appear in both views, or user-side information if any available) in constructing the embedding.

Algorithm 2 outlines the contrastive learning procedure integrated into training. We note that negative sampling can be done implicitly by the denominator summation, which sums over all other samples in the batch (also known as in-batch negatives). This is efficient and commonly used in CL implementations. The above process runs inside each training iteration in parallel with the supervised loss computation described next.

For contrastive learning, we adopt an *in-batch negative sampling* strategy. For each user u , we construct a positive pair $(\mathbf{p}_u, \mathbf{p}'_u)$ consisting of the embeddings from the original and counterfactual graphs. All other users' counterfactual embeddings $\mathbf{p}'_{u'}$ within the same mini-batch (for $u' \neq u$) are treated as negative samples. This approach enables efficient training and naturally provides hard negatives without relying on an external memory bank.

An analogous strategy is applied for item embeddings: $(\mathbf{q}_i, \mathbf{q}'_i)$ forms a positive pair, while $\mathbf{q}'_{i'}$ (for $i' \neq i$) from the same batch serve as negatives. This in-batch sampling

scheme ensures stable convergence and is widely adopted in recent contrastive learning frameworks.

3.6 Recommendation prediction and loss function

For the recommendation task, we use the dot product of user and item embeddings from the original graph as the predictor:

$$\hat{y}_{ui} = \mathbf{p}_u \cdot \mathbf{q}_i = \sum_{f=1}^d p_{u,f} q_{i,f}, \quad (16)$$

where $p_{u,f}$ is the f -th dimension of user u 's embedding and $q_{i,f}$ is the f -th dimension of item i 's embedding.

We adopt the Bayesian Personalized Ranking (BPR) loss for implicit feedback. BPR is a pairwise loss that encourages the score of observed interactions to be higher than that of unobserved ones. For each user u , we sample one positive item i (with $y_{ui} = 1$) and one negative item j (with $y_{uj} = 0$). The loss for this triple (u, i, j) is:

$$\ell_{rec}(u, i, j) = -\ln \sigma(\hat{y}_{ui} - \hat{y}_{uj}), \quad (17)$$

where $\sigma(\cdot)$ is the sigmoid function. This loss is minimized when $\hat{y}_{ui} \gg \hat{y}_{uj}$, i.e., the observed item is scored much higher than the unobserved. In practice, we sum over all sampled triples:

$$\mathcal{L}_{rec} = \sum_{(u,i,j) \in \mathcal{D}} -\ln \sigma(\hat{y}_{ui} - \hat{y}_{uj}), \quad (18)$$

where $\mathcal{D}$ is the set of training triples (we typically generate these by uniform random sampling per epoch). We ensure each user appears in roughly equal number of triples. Other loss functions like pointwise cross-entropy or listwise losses could be used; we chose BPR for its simplicity and good performance in implicit CF tasks.

Finally, the total loss for training is:

$$\mathcal{L} = \mathcal{L}_{rec} + \lambda \mathcal{L}_{CL}, \quad (19)$$

where λ is a hyperparameter. We treat λ as a weighting factor to control the impact of the contrastive objective. A larger λ means we prioritize learning bias-invariant features (possibly at slight expense of fitting the known data), whereas a smaller λ means we focus more on fitting interactions and use CL just lightly as regularization. In our experiments, we tuned λ in 0.1, 0.2, 0.5, 1.0 and found moderate values (0.2-0.5) work well.

Model Training: We train the model using mini-batch stochastic gradient descent (SGD) or Adam optimizer. In each iteration, we sample a batch of users and items, construct the counterfactual graph edges for those (or for the full

Algorithm 3 Training Procedure of COER

Require: Interaction graph $\mathcal{G}$, embedding dim d , layers L , bias drop rate η , CL weight λ , temperature τ , learning rate α

Ensure: Trained embeddings $\mathbf{P}^{(0)}, \mathbf{Q}^{(0)}$

```

1: Initialize user embeddings:  $\mathbf{P}^{(0)} \sim \mathcal{N}(0, 0.1^2)$ 
2: Initialize item embeddings:  $\mathbf{Q}^{(0)} \sim \mathcal{N}(0, 0.1^2)$ 
3: for epoch = 1 to  $N_{\text{epochs}}$  do
4:   Compute item popularity scores  $\{\pi_i\}$  from current  $\mathcal{G}$ 
5:   Construct counterfactual graph  $\mathcal{G}'$  using Algorithm 1 with  $\eta, \{\pi_i\}$ 
6:   for each training batch (sample  $B_U, B_I$ , and for each  $u \in B_U$ 
       sample positive  $i$  and negative  $j$ ) do
7:     Perform  $L$ -layer graph propagation on  $\mathcal{G}$  to get  $\mathbf{p}_u, \mathbf{q}_i$ 
8:     Perform  $L$ -layer graph propagation on  $\mathcal{G}'$  to get  $\mathbf{p}'_u, \mathbf{q}'_i$ 
9:     Compute recommendation loss:

```

$$\mathcal{L}_{\text{rec}} = - \sum_{u \in B_U} \log \sigma(\hat{y}_{ui} - \hat{y}_{uj})$$

```

10:   Compute contrastive loss  $\mathcal{L}_{CL}$  using Algorithm 2
11:   Compute total loss:  $\mathcal{L}_{\text{batch}} = \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{CL}$ 
12:   Update  $\mathbf{P}^{(0)}, \mathbf{Q}^{(0)}$  via gradient descent on  $\mathcal{L}_{\text{batch}}$ 
13: end for
14: end for
15: return  $\mathbf{P}^{(0)}, \mathbf{Q}^{(0)}$ 

```

graph if doing a full pass), compute embeddings, then compute losses as above and backpropagate to update $\mathbf{P}^{(0)}, \mathbf{Q}^{(0)}$. We also update any trainable parameters in the graph propagation (here only the embeddings, since LightGCN has no additional weight matrices). The time complexity per epoch is $O(|E|L)$ for graph convolution (since each edge is used in L layers) plus the contrastive loss computation which is $O((|B_U| + |B_I|)^2)$ for a batch (though implementation can use vectorization). In practice, our method is efficient and scales linearly with interactions, similar to other GCN-based recommenders.

Pseudocode for the entire training process is given in Algorithm 3. It outlines how the components interact each epoch.

3.7 Inference and recommendation generation

After training converges, we use the learned user and item embeddings from the original graph branch (since that encodes the full data). At inference, we do not need to construct a counterfactual graph or do contrastive calculations. We simply obtain $\mathbf{p}_u$ for each target user u and $\mathbf{q}_i$ for each item i . The score for item i relative to user u is $s_{ui} = \mathbf{p}_u \cdot \mathbf{q}_i$. To generate a top- K recommendation list, we compute s_{ui} for all candidate items i not yet interacted by u , then select the K items with highest scores.

As shown in Algorithm 4. This process can be vectorized using matrix operations for efficiency. Let $\mathbf{P} \in \mathbb{R}^{|U| \times d}$ and $\mathbf{Q} \in \mathbb{R}^{|I| \times d}$ be the final embedding matrices. We can compute the score matrix $\hat{\mathbf{Y}} = \mathbf{P}\mathbf{Q}^T$ (of size $|U| \times |I|$). The u -th row of $\hat{\mathbf{Y}}$ contains all item scores for user u . We then mask out

Algorithm 4 Top- K Recommendation Inference

Require: Learned user embeddings $\{\mathbf{p}_u\}$, item embeddings $\{\mathbf{q}_i\}$, candidate item set for each user (excluding seen items), recommendation list length K

Ensure: Top- K recommendation lists for all users

```

1: for each user  $u$  do
2:   for each candidate item  $j$  unseen by  $u$  do
3:     Compute score:  $\hat{y}_{uj} \leftarrow \mathbf{p}_u \cdot \mathbf{q}_j$ 
4:   end for
5:   Rank all candidate items  $j$  by  $\hat{y}_{uj}$  in descending order
6:   Recommend top- $K$  items to user  $u$ 
7: end for
8: return Top- $K$  recommendation lists  $\{\text{Rec}_u\}_u$ 

```

any item that u has interacted with (since recommending an already-taken course is redundant), and pick the top K columns for that row.

The complexity at inference is $O(|U| \cdot |I| \cdot d)$ if done naively, but in practice $|I|$ might be large (e.g., thousands of courses or products). One can leverage efficient libraries or GPUs to compute the score matrix, or apply approximate nearest neighbor search if needed for scalability.

It's worth noting that our model's training objective has made $\mathbf{p}_u$ insensitive to popularity noise. Therefore, the top- K results $\text{topK}(j|\hat{y}_{uj})$ should include relevant niche items that earlier models might miss. We expect the ranking to be more aligned with users' true interests, which is validated by higher precision metrics in our experiments.

4 Experiments

4.1 Datasets

To evaluate the model's applicability across both domain-specific and general recommendation scenarios, we include two educational datasets (XuetangX and Amazon Books) and two widely-used general-purpose benchmarks (MovieLens and TaoBao). The educational datasets exhibit strong domain characteristics, such as course enrollment sparsity, long-tail distribution of learning materials, and popularity-conformity dynamics. In contrast, MovieLens and TaoBao serve as comparative baselines to assess the model's generalizability and robustness across different interaction patterns:

- **MovieLens 1M** (Harper and Konstan 2015): A movie rating dataset with 6,040 users and 3,952 movies, containing 1,000,209 interactions (ratings). We transform ratings into implicit feedback by treating rating ≥ 4 as an interaction (commonly done in literature). This dataset is dense and often used as a benchmarking ground truth for CF models.
- **Amazon Books** (McAuley et al. 2015): A subset of the Amazon product review data for the "Books" category.

It contains 52,643 users, 91,599 items, and 2,984,108 implicit interactions (user purchased or reviewed a book). This dataset is sparse and exhibits strong popularity bias (few bestsellers get disproportionately many interactions). We filtered users with fewer than 5 interactions, as is standard, to focus on users with some history.

- **XuetangX MOOCs** (Yu et al. 2020): A large-scale MOOC (Massive Open Online Courses) dataset from XuetangX, one of the largest online education platforms in China. We use a processed version provided by the MOOCCube dataset. It has 32,223 users and 1,667 courses, with 159,000 enrollment interactions. Each user typically enrolls in only a few courses, and course popularity follows a long-tail distribution. This dataset represents a real educational recommendation scenario, where dropout rates are high and engagement is skewed towards a few popular courses.
- **Taobao E-commerce** (Zhou et al. 2018): A user behavior dataset from Taobao (Alibaba) e-commerce platform, often used for recommender system research. We use a subset of the dataset focused on implicit item clicks. It includes 50,000 users, 36,364 items, and 1,000,000 interactions. While not an educational dataset per se, we include it as a challenging large-scale benchmark. The interactions here might analogously represent, say, engaging with educational materials on a platform like Coursera or an online bookstore scenario, providing diversity in evaluation. Importantly, Taobao data has strong popularity bias and an extremely large item catalogue, testing the model's ability to scale.

For all datasets, we split interactions into training, validation, and test sets. We use a chronological split when timestamps are available (for Amazon and Taobao) or a random 8:1:1 split otherwise (for MovieLens and XuetangX, since they are relatively static). In each test set, for each user we hold out the latest interaction(s) as ground truth and use all prior interactions for training. We ensure each user has at least one training and one test item.

4.2 Evaluation metrics

We adopt Recall@K and NDCG@K (Normalized Discounted Cumulative Gain) as evaluation metrics, which are standard for top- K recommendation (Lin et al. 2021; Zhao et al. 2024). These metrics are defined per user and then averaged. Recall@K measures the proportion of relevant items (e.g., courses the user actually engaged with in a test set) that are present in the top K recommendations:

$$\text{Recall@K}(u) = \frac{|\{\text{relevant items for user } u\} \cap \{\text{top-}K \text{ recommended items for } u\}|}{|\{\text{relevant items for } u\}|}. \quad (20)$$

We report the mean Recall@K across all users. NDCG@K accounts for the rank positions of relevant items, giving higher weight to hits at top ranks. For a single user u , DCG@K is:

$$\text{DCG@K}(u) = \sum_{j=1}^K \frac{2^{\text{rel}(u,j)} - 1}{\log_2(j+1)}, \quad (21)$$

where $\text{rel}(u, j)$ is the relevance of the item at rank j for user u (often binary 0/1 in implicit feedback). The DCG is normalized by the ideal DCG (IDCG) where all relevant items are ranked at the top:

$$\text{NDCG@K}(u) = \frac{\text{DCG@K}(u)}{\text{IDCG@K}(u)}, \quad (22)$$

and we report the average NDCG@K over all users. A higher Recall or NDCG indicates better recommendation performance.

4.3 Hyperparameter tuning details

We conducted systematic hyperparameter tuning using a grid-search strategy on the validation set to ensure fair and reproducible optimization across all datasets. The search ranges, tuning methodology, and rationale for selecting the final optimal values are summarized as follows.

- **Embedding Dimension (d):** We searched $d \in \{32, 64, 128\}$. The value $d = 64$ was selected as it provides the best balance between representation capacity and computational efficiency.
- **Number of GCN Layers (L):** We evaluated $L \in \{1, 2, 3\}$. The setting $L = 2$ consistently achieved the highest validation NDCG, indicating that two-hop neighborhood aggregation is sufficient without introducing oversmoothing.
- **Contrastive Loss Weight (λ):** We tuned $\lambda \in \{0.1, 0.2, 0.3, 0.5, 1.0\}$. Moderate weights yielded the best performance: we used $\lambda = 0.2$ for MovieLens and XuetangX, and $\lambda = 0.3$ for Amazon and Taobao, reflecting the higher sparsity levels of the latter datasets.
- **Intervention Rate (η):** The intervention intensity was selected from $\eta \in \{0.3, 0.5, 0.7\}$. The value $\eta = 0.5$ yielded the most effective reduction of popularity bias while retaining sufficient structural information in the counterfactual graph.
- **Temperature (τ):** We examined $\tau \in \{0.05, 0.1, 0.2\}$. The setting $\tau = 0.1$ provided stable contrastive gradients and consistent performance across datasets.

For tuning methodology, we employed a greedy grid-search strategy: starting from default settings, we tuned one

hyperparameter at a time while fixing the others to the best-performing values found so far. All hyperparameters were validated using a held-out validation set comprising 10% of the training interactions. The complete tuning logs and sensitivity analyses are provided in the supplementary material.

4.4 Baseline models

We compare COER against several state-of-the-art baselines:

- **BPR-MF** (Rendle et al. 2012): Bayesian Personalized Ranking optimized matrix factorization. This is a classic CF model using only dot product latent factors (no graph information), trained with the BPR loss.
- **LightGCN** (He et al. 2020): A simplified graph convolution model that retains only neighborhood aggregation without feature transformations. LightGCN has shown strong performance with fewer parameters and is a core of our model's encoder.
- **SGL-ED** (Qian et al. 2015): This model integrates graph contrastive learning into recommendation. To tackle challenges like long-tail distributions and sparse supervision, SGL introduces contrastive learning by generating multiple views via edge dropout. These views form a three-channel contrastive learning framework, including a main view and two contrastive views. The model jointly optimizes both recommendation loss and contrastive loss in a unified learning paradigm. Among the variants, the ED (Edge Dropout) version delivers the best performance, making it the representative form of the SGL approach.
- **BUIR** (Lee et al. 2021): BUIR is a recommendation framework based solely on first-order collaborative filtering signals. It adopts a two-tower architecture comprising a target network and an online network. Through self-supervised learning, it constructs augmented views for positive sample pairs using neighborhood information, addressing data sparsity. However, the model does not incorporate negative sample pairs in its training, which limits its contrastive learning effectiveness.
- **SimGCL** (Yu et al. 2022): Unlike SGL-ED, SimGCL avoids altering the graph structure. Instead, it injects uniform noise into the embedding space to produce distinct representations of the same node, enabling contrastive learning. This method can be seen as a form of local-global contrastive learning, as it perturbs embeddings without modifying the underlying graph.
- **MFGCL** (Zhao et al. 2023): This is a Top-K recommendation model that leverages context-aware contrastive learning. Built on the LightGCN encoder, it partitions binary user-feature interaction graphs based on node types, extracting representations from multiple perspectives. These are then aggregated to form the

final embedding. The model employs a three-channel contrastive framework, where two of the channels generate perturbed views by injecting random noise into the embedding space, enhancing robustness and representation quality.

- **EduLGCL** (Zhao et al. 2024): The Local-Global Contrastive Learning model for education recommendation. This is the previous state-of-the-art specifically designed for educational data. EduLGCL performs contrastive learning between local context (subgraph around a user) and global context (full graph embeddings) to enhance course recommendations. It also uses the same benchmark datasets in its evaluation, making it a direct point of comparison. EduLGCL does not incorporate causal inference, so its improvements come from advanced contrastive regularization and graph modeling.

For fair comparison, all methods are implemented in the same framework (PyTorch) and optimized with the same training procedure where applicable. We tune hyperparameters for each baseline on the validation set. For example, we set latent dimension $d = 64$ for all models (a common choice balancing accuracy and efficiency), use the optimal learning rate in $1e-4$, $5e-4$, $1e-3$ for each model, and tune any model-specific parameters (e.g., number of layers for GCN models, coefficient of SSL loss for SGL/NCL, etc.). For our COER model, we set $L = 2$ layers (which gave best validation results on most data), $\lambda = 0.2$ for MovieLens/XuetangX and $\lambda = 0.3$ for Amazon/Taobao (slightly more weight on CL for sparser data), $\tau = 0.1$ for the contrastive loss, and $\eta = 0.5$ for the counterfactual graph (meaning we remove on average 50% of edges for the most popular items).

4.5 Main results

We evaluated our proposed COER method against several state-of-the-art recommendation approaches on four benchmark datasets: MovieLens, Amazon, TaoBao, and XuetangX. We employed two common metrics, Recall@K (R@K) and NDCG@K (N@K), with K=10 and K=20. The detailed results are summarized in Tables 1, 2, 3, and 4

Our COER method significantly outperforms all baseline models on MovieLens (Table 1). Specifically, compared with the previously strongest baseline, EduLGCL, COER achieves approximately 1.51% (absolute) increase in Recall@10 and a notable 2.63% (absolute) improvement in NDCG@10, indicating that the causal contrastive learning framework successfully enhances ranking quality and recall rate. At K=20, COER maintains its superior performance, demonstrating robustness in capturing user preferences even when expanding the recommendation list.

On the Amazon dataset (Table 2), which is characterized by high sparsity, COER achieves considerable improvement

Table 1 Results on movielens datasets

Model	Recall@10	NDCG@10	Recall@20	NDCG@20
MF	0.1072	0.2222	0.1782	0.2389
LightGCN	0.1369	0.3382	0.2093	0.3392
LGCACF	0.1169	0.3172	0.1825	0.3447
BUIR	0.1348	0.3201	0.2039	0.3161
SGLED	0.1478	0.3415	0.2339	0.3439
SimGCL	0.1491	0.3429	0.2318	0.3482
MFGCL	0.1466	0.3408	0.2276	0.3391
EduLGCL	0.1572	0.3601	0.2392	0.3562
COER(Ours)	0.1723	0.3864	0.2514	0.3784

Table 2 Results on Amazon Datasets

Model	Recall@10	NDCG@10	Recall@20	NDCG@20
MF	0.0459	0.0535	0.0598	0.0595
LightGCN	0.0568	0.0592	0.0810	0.0698
LGCACF	0.0636	0.0750	0.0851	0.0844
BUIR	0.0355	0.0362	0.0572	0.0461
SGLED	0.0709	0.0825	0.0932	0.0917
SimGCL	0.0685	0.0794	0.0911	0.0886
MFGCL	0.0667	0.0738	0.0915	0.0852
EduLGCL	0.0748	0.0845	0.0986	0.0943
COER(Ours)	0.0974	0.0988	0.0997	0.0994

Table 3 Results on TaoBao Datasets

Model	Recall@10	NDCG@10	Recall@20	NDCG@20
MF	0.0478	0.0387	0.0579	0.0422
LightGCN	0.0479	0.0343	0.0710	0.0419
BUIR	0.0310	0.0201	0.0483	0.0267
SGLED	0.0590	0.0422	0.0865	0.0515
SimGCL	0.0583	0.0420	0.0850	0.0509
MFGCL	0.0577	0.0412	0.0864	0.0510
EduLGCL	0.0589	0.0423	0.0865	0.0514
COER(Ours)	0.0791	0.0643	0.0996	0.0723

Table 4 Results on XuetangX Datasets

Model	Recall@10	NDCG@10	Recall@20	NDCG@20
MF	0.4781	0.3861	0.5797	0.4213
LightGCN	0.5210	0.4476	0.6158	0.4796
BUIR	0.4394	0.3945	0.5546	0.4311
SGLED	0.5416	0.4710	0.6481	0.5050
SimGCL	0.5382	0.4730	0.6374	0.5070
MFGCL	0.5312	0.4527	0.6287	0.4869
EduLGCL	0.5437	0.4701	0.6431	0.5038
COER(Ours)	0.5546	0.4816	0.6674	0.5264

over existing methods, particularly surpassing EduLGCL by 2.26% in Recall@10 and 1.43% in NDCG@10. The substantial margin highlights COER's ability to better exploit sparse interactions through its causal debiasing strategy, effectively alleviating popularity bias that is notably prevalent in Amazon's dataset. Interestingly, at Recall@20, COER and EduLGCL perform similarly, but COER maintains a clear advantage in ranking quality (NDCG@20), suggesting superior relevance ordering even when more items are recommended.

The TaoBao dataset (Table 3), characterized by heavy popularity bias and very sparse interactions, further validates COER's robustness and effectiveness. COER notably surpasses EduLGCL by approximately 2.02% in Recall@10 and 2.20% in NDCG@10. At K=20, COER continues to lead, confirming that our causal contrastive design particularly benefits scenarios with challenging long-tail distributions by accurately capturing user genuine interests rather than popularity-driven interactions.

In the education-specific XuetangX dataset (Table 4), COER demonstrates clear superiority, improving upon EduLGCL by approximately 1.09% in Recall@10 and 1.15% in NDCG@10. This indicates the effectiveness of our causal contrastive learning approach, explicitly tailored for educational contexts to mitigate bias effects such as course popularity or conformity.

The experimental analyses collectively demonstrate that our COER method consistently achieves state-of-the-art performance across all benchmark datasets, particularly in settings characterized by pronounced sparsity and strong popularity bias. Beyond the numerical gains, the results reveal several important insights. First, COER's causal graph augmentation effectively suppresses spurious correlations caused by over-exposure to popular items, enabling the model to identify user-item matches that better reflect intrinsic interests rather than platform-driven popularity. Second, the contrastive alignment between factual and counterfactual views reinforces stability in the learned representations, leading to more reliable ranking behavior even when user histories are extremely limited. These mechanisms jointly contribute to substantial improvements in both recall and ranking quality, confirming the advantage of integrating causal intervention with contrastive representation learning for educational recommendation scenarios.

Table 5 Performance comparison across different COER variants on Movielens and Amazon datasets

Variant	Movielens R@10	Movielens N@10	Amazon R@10	Amazon N@10
M1	0.1598	0.3653	0.0795	0.0862
M2	0.1382	0.3394	0.0592	0.0623
M3	0.1645	0.3709	0.0847	0.0885
Full COER	0.1723	0.3864	0.0974	0.0988

4.6 Ablation study

To systematically evaluate the contribution of each component within our proposed COER model, we conducted detailed ablation experiments. Specifically, we explored three ablation scenarios:

- **M1 (w/o Causal Augmentation):** Removing the causal augmentation module and only applying random graph augmentations (random dropout).
- **M2 (w/o Contrastive Loss):** Removing the contrastive learning module, thus reducing COER to a pure LightGCN model.
- **M3 (Random Sampling Instead of Popularity-Based Sampling):** Replacing the popularity-based edge sampling strategy with purely random edge sampling.
- **Full COER:** The complete proposed model.

The results on representative datasets (Movielens and Amazon) are shown in Table 5:

Removing the causal augmentation module (M1) leads to a significant performance decrease, demonstrating the importance of explicitly modeling causal relations to effectively mitigate popularity biases.

Eliminating the contrastive loss module (M2) results in performance nearly identical to LightGCN, highlighting the crucial role of contrastive learning in producing robust and informative embeddings.

Replacing popularity-based sampling with random sampling (M3) also negatively impacts performance, though not as severely as M1 or M2, suggesting that while random augmentation can improve performance slightly, targeted causal interventions more effectively mitigate popularity biases.

The complete COER model consistently outperforms all ablated variants, confirming the synergy and necessity of each component.

4.7 Popularity bias mitigation analysis

To explore how effectively COER mitigates popularity bias, we divided the items in the Movielens and Amazon datasets into three popularity quantiles: low, medium, and high popularity. We then evaluated Recall@10 for each group. The results are summarized in Table 6:

Table 6 Performance comparison across popularity levels for COER and EduLGCL on Movielens and Amazon datasets

Popularity	Movielens-COER	Movielens-EduLGCL	Amazon-COER	Amazon-EduLGCL
Low	0.158	0.134	0.062	0.043
Medium	0.175	0.160	0.102	0.081
High	0.189	0.191	0.128	0.120

COER significantly improves recommendations for low- and medium-popularity items compared to EduLGCL, particularly in the long-tail scenario. For low-popularity items, the relative improvement is approximately 2.4

Both methods perform comparably for high-popularity items, indicating that the primary advantage of COER lies in effectively addressing popularity bias, ensuring diverse and fair recommendations, especially for less popular items.

4.8 Robustness to data sparsity analysis

To evaluate the robustness of COER under varying data sparsity conditions, we categorized users in the Movielens and TaoBao datasets based on the number of historical interactions:

- Sparse users (fewer than 10 interactions)
- Moderately sparse users (10 to 30 interactions)
- Dense users (more than 30 interactions)

The Recall@10 results are presented in Table 7:

COER consistently outperforms EduLGCL across all sparsity levels, demonstrating strong robustness and effective utilization of limited interaction data through its causal and contrastive learning components.

The improvement is particularly pronounced for sparse users, with approximately 1.6% higher Recall@10 on Movielens and 1.5% on TaoBao, clearly indicating that COER effectively alleviates data sparsity challenges.

Combining the insights from the ablation and additional experiments, we have rigorously validated the effectiveness of our COER model:

- Ablation Study conclusively demonstrated the importance of each proposed component (causal augmentation, contrastive learning, and popularity-based sampling),

highlighting their combined contribution to superior recommendation performance.

- Additional Experiments further validated COER's superior ability to mitigate popularity bias and effectively address data sparsity. These results underscore the generalization capability and robustness of the proposed causal contrastive learning framework.

Overall, COER provides a principled solution to critical challenges in educational recommendation scenarios, demonstrating substantial theoretical significance and promising practical application potential.

4.9 Case study: XuetangX educational data analysis

To further investigate the domain adaptability of our proposed model, we conduct a dedicated analysis on XuetangX, a large-scale educational dataset reflecting realistic characteristics of Massive Open Online Course (MOOC) platforms. XuetangX presents specific challenges, including data sparsity, skewed popularity distribution, and cross-disciplinary learner interests, all of which distinguish educational recommendation from general product recommendation.

Cold-start learner robustness. A key challenge in MOOCs is that many learners interact with only a handful of courses. We isolate users with fewer than 5 interactions (approx. 41% of the dataset) and evaluate top-10 recall performance. Our model COER achieves a 5.2% improvement over EduLGCL and 7.1% over SimGCL, showing its strength in sparse settings common in educational platforms.

Recovery of niche course interests. We partition courses into three popularity bands. In the long-tail group (lowest 30% by enrollment), COER's top-10 recommendations include 34% from this group, while EduLGCL covers only 19%. This indicates that COER not only mitigates popularity bias but is more effective in surfacing niche courses relevant to individual learning needs.

Table 7 Performance under varying user sparsity levels for COER and EduLGCL on Movielens and TaoBao datasets

User Sparsity	Movielens-COER	Movielens-EduLGCL	TaoBao-COER	TaoBao-EduLGCL
Sparse users	0.112	0.096	0.056	0.041
Medium users	0.169	0.151	0.081	0.062
Dense users	0.221	0.210	0.112	0.101

Interdisciplinary recommendation diversity. We further analyze recommendation diversity across disciplines (e.g., science, humanities, engineering, arts). COER recommends at least one course from a distinct secondary domain in 57% of users, compared to 42% in EduLGCL, showing a stronger alignment with cross-interest learners and curriculum exploration goals.

Learning path continuity. For users who took a course in domain A (e.g., ?Introduction to Psychology?), COER is more likely to recommend semantically relevant but deeper or adjacent courses in the same field (e.g., ?Cognitive Science? or ?Behavioral Neuroscience?), whereas baselines tend to over-recommend trending or general courses. This suggests that COER can support personalized learning trajectories.

These findings confirm that COER offers enhanced educational value beyond accuracy: it enables equitable exposure to diverse and meaningful learning opportunities, aligns with pedagogical continuity, and supports personalized exploration in large-scale open education systems.

5 Conclusion

In this paper, we proposed COER, a novel educational recommendation model integrating causal inference with contrastive learning. By generating a counterfactual interaction graph that reduces popularity bias and employing contrastive representation learning, COER learns user-item embeddings robust to confounding popularity effects. Extensive experiments on Movielens, Amazon, XuetaangX, and TaoBao datasets demonstrated that COER significantly surpasses previous state-of-the-art methods, particularly benefiting long-tail recommendations and sparse user scenarios. Ablation studies highlighted the necessity of each component, affirming our approach's effectiveness. Future research directions include extending the framework with content-based causal augmentations, adaptive causal sampling strategies, online evaluations, and exploring causal explainability for trustworthy recommendations.

While our approach demonstrates strong performance in educational recommendation tasks, it does rely on specific hyperparameters such as the popularity threshold η in the counterfactual graph construction. These thresholds may require domain-specific tuning, which could limit the model's direct transferability. Moreover, while we evaluated COER on general and educational datasets, further validation in other domains (e.g., news, e-commerce) is needed to assess broader applicability. In future work, we plan to explore adaptive methods for causal intervention strength, incorporate content-based features into the causal graph, and investigate real-time recommendation in online learning environments.

Data Availability The datasets used in this manuscript are publicly available datasets. Detailed information about these datasets is provided in Section Dataset of this manuscript.

Declarations

Competing Interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Credit Author Statement Yan Zhou: Methodology, Writing-original draft, Software, Formal analysis, Resources, Project administration. Yanguang Chen: Methodology, Supervision, Resources, Funding acquisition, Writing-review & editing.

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